Analytical Laboratory Report

Report ID: S59163.01(01)
Generated on 03/18/2024

Report to

Attention: Cindy Tomaszewski
City of Cadillac - WWTP
1121 Plett Road
Cadillac, MI 49601

Phone: 231-775-2368 FAX:
Email: lab@cadillac-mi.net

Report produced by

Merit Laboratories, Inc.
2680 East Lansing Drive
East Lansing, MI 48823

Phone: (517) 332-0167 FAX: (517) 332-6333

Contacts for report questions:
John Laverty (johnlaverty@meritlabs.com)
Barbara Ball (bball@meritlabs.com)

Report Summary

Lab Sample ID(s): S59163.01-S59163.02
Project: Qtrly Biosolids
Collected Date(s): 02/27/2024
Submitted Date/Time: 02/28/2024 10:20
Sampled by: David Skiff
P.O. #:

Table of Contents

Maya Murshak
Technical Director

General Report Notes

Analytical results relate only to the samples tested, in the condition received by the laboratory.
Methods may be modified for improved performance.
Results reported on a dry weight basis where applicable.
'Not detected' indicates that parameter was not found at a level equal to or greater than the reporting limit (RL).
When MDL results are provided, then 'Not detected' indicates that parameter was not found at a level equal to or greater than the MDL.
40 CFR Part 136 Table II Required Containers, Preservation Techniques and Holding Times for the Clean Water Act specify that samples for acrolein and acrylonitrile, and 2-chloroethylvinyl ether need to be preserved at a pH in the range of 4 to 5 or if not preserved, analyzed within 3 days of sampling.
QA/QC corresponding to this analytical report is a separate document with the same Merit ID reference and is available upon request.
Starred (*) analytes are not NY NELAP accredited.
Samples are held by the lab for 30 days from the final report date unless a written request to hold longer is provided by the client.
Report shall not be reproduced except in full, without the written approval of Merit Laboratories, Inc.
Limits for drinking water samples, are listed as the MCL Limits (Maximum Contaminant Level Concentrations)
PFAS requirement: Section 9.3.8 of U.S. EPA Method 537.1 states "If the method analyte(s) found in the Field Sample is present in the FRB at a concentration greater than 1/3 the MRL, then all samples collected with that FRB are invalid and must be recollected and reanalyzed."
Samples submitted without an accompanying FRB may not be acceptable for compliance purposes.
Wisconsin PFAs analysis: MDL = LOD; RL = LOQ. LOD and LOQ are adjusted for dilution.
All accreditations/certifications held by this laboratory are listed on page 3. Not all accreditations/certifications are applicable to this report.
For a specific list of accredited analytes, please feel free to contact the laboratory or visit https://www.meritlabs.com/certifications.

Report Narrative

There is no additional narrative for this analytical report

Laboratory Accreditations (For Reference Only)

Authority	Accreditation ID
Michigan DEQ	#9956
DOD ELAP & ISO/IEC 17025:2017	#69699 PJLA Testing
WBENC	#2005110032
Ohio VAP	#CL0002
Indiana DOH	#C-MI-07
New York NELAC	#11814
North Carolina DENR	#680
North Carolina DOH	#26702
Pennsylvania DEP	#68-05884
Wisconsin DNR	FID# 399147320

Qualifier Descriptions

Qualifier	Description
!	Result is outside of stated limit criteria
B	Compound also found in associated method blank
E	Concentration exceeds calibration range
F	Analysis run outside of holding time
G	Estimated result due to extraction run outside of holding time
H	Sample submitted and run outside of holding time
I	Matrix interference with internal standard
J	Estimated value less than reporting limit, but greater than MDL
L	Elevated reporting limit due to low sample amount
M	Result reported to MDL not RDL
O	Analysis performed by outside laboratory. See attached report.
R	Preliminary result
S	Surrogate recovery outside of control limits
T	No correction for total solids
X	Elevated reporting limit due to matrix interference
Y	Elevated reporting limit due to high target concentration
b	Value detected less than reporting limit, but greater than MDL
e	Reported value estimated due to interference
j	Analyte also found in associated method blank
p	Benzo(b)Fluoranthene and Benzo(k)Fluoranthene integrated as one peak.
x	Preserved from bulk sample

Glossary of Abbreviations

Abbreviation	Description
RL/RDL	Reporting Limit
MDL	Method Detection Limit
MS	Matrix Spike
MSD	Matrix Spike Duplicate
SW	EPA SW 846 (Soil and Wastewater) Methods
E	EPA Methods
SM	Standard Methods
LN	Linear
BR	Branched

Method Summary

Method	Version
ASTM D7968-17M	ASTM Method D7968 - 17 Modified (Isotopic Dilution)
ASTMD7979-19M	ASTM Method D7979 - 19 Modified (Isotopic Dilution)
SM2540B	Standard Method 2540 B 2015

Parameter Summary

Parameter	Synonym	Cas #
PFBA	Perfluorobutanoic Acid	375-22-4
PFPeA	Perfluoropentanoic Acid	2706-90-3
4:2 FTSA	4:2 Fluorotelomer Sulfonic Acid	757124-72-4
PFHxA	Perfluorohexanoic Acid	307-24-4
PFBS	Perfluorobutane sulfonic Acid	375-73-5
PFHpA	Perfluoroheptanoic Acid	375-85-9
PFPeS	Perfluoropentane Sulfonic Acid	2706-91-4
6:2 FTSA	6:2 Fluorotelomer Sulfonic Acid	27619-97-2
PFOA	Perfluorooctanoic Acid	335-67-1
PFHxS	Perfluorohexane Sulfonic Acid	355-46-4
PFHxS-LN	Perfluorohexane Sulfonic Acid - LN	355-46-4-LN
PFHxS-BR	Perfluorohexane Sulfonic Acid - BR	355-46-4-BR
PFNA	Perfluorononanoic Acid	375-95-1
8:2 FTSA	8:2 Fluorotelomer Sulfonic Acid	39108-34-4
PFHpS	Perfluoroheptane Sulfonic Acid	375-92-8
PFDA	Perfluorodecanoic Acid	335-76-2
N-MeFOSAA	N-methyl perfluorooctanesulfonamidoacetic acid	2355-31-9
EtFOSAA	N-Ethyl Perfluorooctane Sulfonamidoacetic Acid	2991-50-6
PFOS	Perfluorooctane Sulfonic Acid	1763-23-1
PFOS-LN	Perfluorooctane Sulfonic Acid - LN	1763-23-1-LN
PFOS-BR	Perfluorooctane Sulfonic Acid - BR	1763-23-1-BR
PFUnDA	Perfluoroundecanoic Acid	2058-94-8
PFNS	Perfluorononane Sulfonic Acid	68259-12-1
PFDoDA	Perfluorododecanoic Acid	307-55-1
PFDS	Perfluorodecane Sulfonic Acid	335-77-3
PFTrDA	Perfluorotridecanoic Acid	72629-94-8
FOSA	Perfluorooctane Sulfonamide	754-91-6
PFTeDA	Perfluorotetradecanoic Acid	376-06-7
11Cl-PF3OUdS	11-chloroeicosafluoro-3-oxaundecane-1-sulfonic acid	763051-92-9
9Cl-PF3ONS	9-chlorohexadecafluoro-3-oxanone1-sulfonic acid	756426-58-1
ADONA	4,8-dioxa-3H-perfluorononanoic acid	919005-14-4
HFPO-DA	Hexafluoropropylene oxide dimer	13252-13-6
FHpPA (7:3 FTCA)	3-Perfluoroheptyl propanoic acid	812-70-4
FPePA (5:3 FTCA)	3-Perfluoropentyl propanoic acid	914637-49-3
FPrPA (3:3 FTCA)	3-Perfluoropropyl propanoic acid	356-02-5
PFBSA	Perfluorobutanesulfonamide	30334-69-1
PFECHS	Perfluoro-4-ethylcyclohexanesulfonate	67584-42-3
PFHxSA	Perfluorohexanesulfonamide	41997-13-1

Sample Summary (2 samples)

Sample ID	Sample Tag	Matrix	Collected Date/Time
S59163.01	Equipment Blank	Water	02/27/24 13:20
S59163.02	Biosolids - DIG.4	Sludge	02/27/24 13:20

Merit Laboratories, Inc.

Analytical Laboratory Report

Lab Sample ID: S59163.01
Sample Tag: Equipment Blank
Collected Date/Time: 02/27/2024 13:20
Matrix: Water
COC Reference: 159864

Sample Containers

#	Type	Preservative(s)	Refrigerated?	Arrival Temp. (C)	Thermometer #
1	15mL Centrifuge Tube	Trizma	Yes	6.0	IR

Extraction / Prep.

Parameter	Result	Method	Run Date	Analyst	Flags
Initial wt. (g) / Final wt. (g) / Volume (ml)*	12.22/6.53/10	ASTMD7979-19M	02/29/24 12:00	SRP	

Organics

34 PFAs, Method: ASTMD7979-19M, Run Date: 02/29/24 23:36, Analyst: KCV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
PFBA*	Not detected	0.0088		ug/L	1.76	375-22-4	
PFPeA*	Not detected	0.0035		ug/L	1.76	2706-90-3	
4:2 FTSA*	Not detected	0.0018		ug/L	1.76	757124-72-4	
PFHxA*	Not detected	0.0018		ug/L	1.76	307-24-4	
PFBS*	Not detected	0.0018		ug/L	1.76	375-73-5	
PFHpA*	Not detected	0.0018		ug/L	1.76	375-85-9	
PFPeS*	Not detected	0.0018		ug/L	1.76	2706-91-4	
6:2 FTSA*	Not detected	0.0018		ug/L	1.76	27619-97-2	
PFOA*	Not detected	0.0018		ug/L	1.76	335-67-1	
PFHxS*	Not detected	0.0018		ug/L	1.76	355-46-4	
PFHxS-LN*	Not detected	0.0018		ug/L	1.76	355-46-4-LN	
PFHxS-BR*	Not detected	0.0018		ug/L	1.76	355-46-4-BR	
PFNA*	Not detected	0.0018		ug/L	1.76	375-95-1	
8:2 FTSA*	Not detected	0.0018		ug/L	1.76	39108-34-4	
PFHpS*	Not detected	0.0018		ug/L	1.76	375-92-8	
PFDA*	Not detected	0.0018		ug/L	1.76	335-76-2	
N-MeFOSAA*	Not detected	0.0018		ug/L	1.76	2355-31-9	
EtFOSAA*	Not detected	0.0035		ug/L	1.76	2991-50-6	
PFOS*	Not detected	0.0018		ug/L	1.76	1763-23-1	
PFOS-LN*	Not detected	0.0018		ug/L	1.76	1763-23-1-LN	
PFOS-BR*	Not detected	0.0018		ug/L	1.76	1763-23-1-BR	
PFUnDA*	Not detected	0.0018		ug/L	1.76	2058-94-8	
PFNS*	Not detected	0.0018		ug/L	1.76	68259-12-1	
PFDoDA*	Not detected	0.0018		ug/L	1.76	307-55-1	
PFDS*	Not detected	0.0018		ug/L	1.76	335-77-3	
PFTrDA*	Not detected	0.0018		ug/L	1.76	72629-94-8	
FOSA*	Not detected	0.0018		ug/L	1.76	754-91-6	
PFTeDA*	Not detected	0.0035		ug/L	1.76	376-06-7	
11Cl-PF3OUdS*	Not detected	0.0018		ug/L	1.76	763051-92-9	
9Cl-PF3ONS*	Not detected	0.0018		ug/L	1.76	756426-58-1	
ADONA*	Not detected	0.0018		ug/L	1.76	919005-14-4	
HFPO-DA*	Not detected	0.0088		ug/L	1.76	13252-13-6	
FHpPA (7:3 FTCA)*	Not detected	0.0088		ug/L	1.76	812-70-4	
FPePA (5:3 FTCA)*	Not detected	0.0088		ug/L	1.76	914637-49-3	
FPrPA (3:3 FTCA)*	Not detected	0.0088		ug/L	1.76	356-02-5	
PFBSA*	Not detected	0.0018		ug/L	1.76	30334-69-1	
PFECHS*	Not detected	0.0018		ug/L	1.76	67584-42-3	

Lab Sample ID: S59163.01 (continued)
Sample Tag: Equipment Blank

34 PFAs, Method: ASTMD7979-19M, Run Date: 02/29/24 23:36, Analyst: KCV (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
PFHxSA*	Not detected	0.0018		ug/L	1.76	41997-13-1	

Analytical Laboratory Report

Lab Sample ID: S59163.02
Sample Tag: Biosolids - DIG.4
Collected Date/Time: 02/27/2024 13:20
Matrix: Sludge
COC Reference: 159864

Sample Containers

#	Type	Preservative(s)	Refrigerated?	Arrival Temp. (C)	Thermometer #
1	15mL Centrifuge Tube	Trizma	Yes	6.0	IR
1	125mL Plastic	None	Yes	6.0	IR

Extraction / Prep.

Parameter	Result	Method	Run Date	Analyst	Flags
Initial wt. (g) / Final wt. (g) / Volume (ml)*	8.80/6.54/10	ASTM D7968-17M	03/05/24 11:00	SRP	

Inorganics

Method: SM2540B, Run Date: 02/28/24 12:39, Analyst: MAM

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Total Solids*	2.6	1		%	1		

Organics

34 PFAs, Method: ASTM D7968-17M, Run Date: 03/06/24 00:33, Analyst: KCV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
PFBA*	Not detected	3.4		ug/kg	170	375-22-4	
PFPeA*	Not detected	1.7		ug/kg	170	2706-90-3	
4:2 FTSA*	Not detected	1.7		ug/kg	170	757124-72-4	I
PFHxA*	Not detected	1.7		ug/kg	170	307-24-4	
PFBS*	Not detected	1.7		ug/kg	170	375-73-5	
PFHpA*	Not detected	1.7		ug/kg	170	375-85-9	
PFPeS*	Not detected	1.7		ug/kg	170	2706-91-4	
6:2 FTSA*	Not detected	1.7		ug/kg	170	27619-97-2	I
PFOA*	Not detected	1.7		ug/kg	170	335-67-1	
PFHxS*	Not detected	1.7		ug/kg	170	355-46-4	
PFHxS-LN*	Not detected	1.7		ug/kg	170	355-46-4-LN	
PFHxS-BR*	Not detected	1.7		ug/kg	170	355-46-4-BR	
PFNA*	Not detected	1.7		ug/kg	170	375-95-1	
8:2 FTSA*	Not detected	1.7		ug/kg	170	39108-34-4	I
PFHpS*	Not detected	1.7		ug/kg	170	375-92-8	
PFDA*	Not detected	1.7		ug/kg	170	335-76-2	
N-MeFOSAA*	7.7	1.7		ug/kg	170	2355-31-9	
EtFOSAA*	2.1	1.7		ug/kg	170	2991-50-6	I
PFOS*	Not detected	1.7		ug/kg	170	1763-23-1	
PFOS-LN*	Not detected	1.7		ug/kg	170	1763-23-1-LN	
PFOS-BR*	Not detected	1.7		ug/kg	170	1763-23-1-BR	
PFUnDA*	Not detected	1.7		ug/kg	170	2058-94-8	
PFNS*	Not detected	1.7		ug/kg	170	68259-12-1	
PFDoDA*	Not detected	1.7		ug/kg	170	307-55-1	I
PFDS*	Not detected	1.7		ug/kg	170	335-77-3	
PFTrDA*	Not detected	1.7		ug/kg	170	72629-94-8	I
FOSA*	Not detected	1.7		ug/kg	170	754-91-6	
PFTeDA*	Not detected	1.7		ug/kg	170	376-06-7	
11Cl-PF3OUdS*	Not detected	1.7		ug/kg	170	763051-92-9	
9Cl-PF3ONS*	Not detected	1.7		ug/kg	170	756426-58-1	

I-Matrix interference with internal standard

Lab Sample ID: S59163.02 (continued)
Sample Tag: Biosolids - DIG.4

34 PFAs, Method: ASTM D7968-17M, Run Date: 03/06/24 00:33, Analyst: KCV (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
ADONA*	Not detected	1.7		ug/kg	170	919005-14-4	
HFPO-DA*	Not detected	1.7		ug/kg	170	13252-13-6	
FHpPA (7:3 FTCA)*	13	1.7		ug/kg	170	812-70-4	
FPePA (5:3 FTCA)*	87	1.7		ug/kg	170	914637-49-3	
FPrPA (3:3 FTCA)*	Not detected	1.7		ug/kg	170	356-02-5	
PFBSA*	Not detected	1.7		ug/kg	170	30334-69-1	
PFECHS*	Not detected	1.7		ug/kg	170	67584-42-3	
PFHxSA*	Not detected	1.7		ug/kg	170	41997-13-1	

Merit Laboratories Login Checklist

Lab Set ID: S59163
Client: MISCPFC (City of Cadillac - WWTP)
Project: Qtrly Biosolids
Submitted: 02/28/2024 10:20   Login User: MMC

Attention: Cindy Tomaszewski
Address: City of Cadillac - WWTP
1121 Plett Road
Cadillac, MI 49601

Phone: 231-775-2368   FAX:
Email: lab@cadillac-mi.net

Selection				Description	Note
Sample Receiving					
01.	[X] Yes	[ ] No	[ ] N/A	Samples are received at 4C +/- 2C Thermometer #	IR 6.0
02.	[X] Yes	[ ] No	[ ] N/A	Received on ice/ cooling process begun	
03.	[X] Yes	[ ] No	[ ] N/A	Samples shipped	UPS
04.	[ ] Yes	[X] No	[ ] N/A	Samples left in 24 hr. drop box	
05.	[X] Yes	[ ] No	[ ] N/A	Are there custody seals/tape or is the drop box locked	
Chain of Custody					
06.	[X] Yes	[ ] No	[ ] N/A	COC adequately filled out	
07.	[X] Yes	[ ] No	[ ] N/A	COC signed and relinquished to the lab	
08.	[X] Yes	[ ] No	[ ] N/A	Sample tag on bottles match COC	
09.	[ ] Yes	[X] No	[ ] N/A	Subcontracting needed? Subcontacted to:	
Preservation					
10.	[X] Yes	[ ] No	[ ] N/A	Do sample have correct chemical preservation	
11.	[ ] Yes	[ ] No	[X] N/A	Completed pH checks on preserved samples? (no VOAs)	
12.	[ ] Yes	[X] No	[ ] N/A	Did any samples need to be preserved in the lab?	
Bottle Conditions					
13.	[X] Yes	[ ] No	[ ] N/A	All bottles intact	
14.	[X] Yes	[ ] No	[ ] N/A	Appropriate analytical bottles are used	
15.	[X] Yes	[ ] No	[ ] N/A	Merit bottles used	
16.	[X] Yes	[ ] No	[ ] N/A	Sufficient sample volume received	
17.	[ ] Yes	[X] No	[ ] N/A	Samples require laboratory filtration	
18.	[X] Yes	[ ] No	[ ] N/A	Samples submitted within holding time	
19.	[ ] Yes	[ ] No	[X] N/A	Do water VOC or TOX bottles contain headspace	

Corrective action for all exceptions is to call the client and to notify the project manager.

Client Review By: ______________________   Date: __________

Merit Laboratories, Inc.

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Phone (517) 332-0167 Fax (517) 332-4034
www.meritlabs.com

C.O.C. PAGE # 1 OF 1

159864

CHAIN OF COUNTODY RECORD

REPORT TO

CONTACT NAME: CINDY TOMASZEWSKI
COMPANY: CITY OF CADILLAC - WWTP
ADDRESS: 1121 PLETT RD
CITY: CADILLAC MI STATE: MI ZIP CODE: 49601
PHONE NO.: 231-775-2368 CELL NO.: P.O. NO.:
E-MAIL ADDRESS: lab@cadillac-mi.net QUOTE NO.:

INVOICE TO

CONTACT NAME: [X] SAME
COMPANY:
ADDRESS:
CITY: STATE: ZIP CODE:
PHONE NO.: E-MAIL ADDRESS:

PROJECT NO./NAME: QTRLY BIOSOLIDS
SAMPLER(S) - PLEASE PRINT/SIGN NAME: DAVID SKIFF

TURNAROUND TIME REQUIRED ☐ 1 DAY ☐ 2 DAYS ☐ 3 DAYS [X] STANDARD ☐ OTHER

DELIVERABLES REQUIRED ☐ STD [X] LEVEL II ☐ LEVEL III ☐ LEVEL IV ☐ EDD ☐ OTHER

MATRIX CODE: W=WATER GW=GROUNDWATER WW=WASTEWATER S=SOIL L=LIQUID SD=SOLID SL=SLUDGE DW=DRINKING WATER O=OIL WP=WIPE A=AIR WS=WASTE

ANALYSIS (ATTACH LIST IF MORE SPACE IS REQUIRED)

Certifications: ☐ OHIO VAP ☐ Drinking Water ☐ DoD [X] NPDES
Project Locations: ☐ Detroit ☐ New York ☐ Other

MERIT LAB NO. (FOR LAB USE ONLY)	COLLECTION DATE	COLLECTION TIME	SAMPLE TAG IDENTIFICATION-DESCRIPTION	MATRIX	# OF BOTTLES	NONE	HCl	HNO_3	H_2SO_4	NaOH	MeOH	OTHER	PFAS-MI	Special Instructions
59163.01	2-27-24	1320	EQUIPMENT BLANK	W	3	X							X	
.02	2-27-24	1320	BIOSOLIDS - DIG. 4	SL	4	X							X	ASTM D7968

RELINQUISHED BY: SIGNATURE/ORGANIZATION	CA Tomaszewski / City of Cadillac ☐ Sampler	DATE 2/27	TIME 1410	
RECEIVED BY: SIGNATURE/ORGANIZATION	Ship via UPS Ground	DATE	TIME	
RELINQUISHED BY: SIGNATURE/ORGANIZATION	UPS	DATE 2/28/24	TIME 1020	
RECEIVED BY: SIGNATURE/ORGANIZATION	M Chilcote	DATE 2/28/24	TIME 1020	
RELINQUISHED BY: SIGNATURE/ORGANIZATION		DATE	TIME	
RECEIVED BY: SIGNATURE/ORGANIZATION		DATE	TIME	

SEAL NO. SEAL INTACT YES☐ NO☐ INITIALS
SEAL NO. SEAL INTACT YES☐ NO☐ INITIALS

NOTES: TEMP. ON ARRIVAL 6.0

PLEASE NOTE: SIGNING ACKNOWLEDGES ADHERENCE TO MERIT'S SAMPLE ACCEPTANCE POLICY ON REVERSE SIDE

Rev. 1.26.22

Quality Control Report

Report ID: QC-S59163-01
Generated on 03/18/2024

Report to

Attention: Cindy Tomaszewski
City of Cadillac - WWTP
1121 Plett Road
Cadillac, MI 49601

Phone: 231-775-2368 FAX:

Report Produced by

Merit Laboratories
2680 East Lansing Drive
East Lansing, MI 48823

Phone: (517) 332-0167 FAX: (517) 332-6333

Report Summary

Lab Sample ID(s): S59163.01-S59163.02
Project: Qtrly Biosolids
Submitted Date/Time: 02/28/2024 10:20
Sampled by: David Skiff
P.O. #:

QC Report Sections

Report Flag Descriptions

*: QC result is outside of indicated control limits
W: Surrogate result not applicable due to sample dilution

I certify that this data package is in compliance with the terms and conditions of the program, and project, and contractual requirements both technically and for completeness. Release of the data contained in this hardcopy data package and its computer-readable data submitted has been authorized by the Quality Assurance Manager and his/her designee, as verified by the following signature.

Barbara Ball

Barbara Ball
Quality Assurance Manager

Lab Sample ID: S59163.01
Sample Tag: Equipment Blank
Collected Date/Time: 02/27/2024 13:20
Matrix: Water
COC Reference: 159864

Analysis	Method	Run Date/Time	Batch ID	Prep ID	Surr	QC Types
Organics - Volatiles						
34 PFAs	ASTMD7979-19M	02/29/24 23:36	AK240229	PF240229W1	Yes	BLK/LCS/LCSD/MS/DU

QC Report - Analysis Summary

Lab Sample ID: S59163.02

Sample Tag: Biosolids - DIG.4

Collected Date/Time: 02/27/2024 13:20

Matrix: Sludge

COC Reference: 159864

Analysis	Method	Run Date/Time	Batch ID	Prep ID	Surr	QC Types
Inorganics						
Total Solids	SM2540B	02/28/24 12:39	TS240228A	TS240228A	No	BLK/LCS/DUP
Organics - Volatiles						
34 PFAs	ASTM D7968-17M	03/06/24 00:33	AK240305	PF240305S1	Yes	BLK/LCS/LCSD/MS/MS

QC Report - Prep Batch Summary

Inorganics, Prep Batch ID: TS240228A

Surrogates: No, QC Types: BLK/LCS/DUP

Sample ID	Analysis	Method	Run Date/Time	Batch ID
S59163.02	Total Solids	SM2540B	02/28/24 12:39	TS240228A

Organics - Volatiles, Prep Batch ID: PF240229W1

Surrogates: Yes, QC Types: BLK/LCS/LCSD/MS/DUP

Sample ID	Analysis	Method	Run Date/Time	Batch ID
S59163.01	34 PFAs	ASTMD7979-19M	02/29/24 23:36	AK240229

Organics - Volatiles, Prep Batch ID: PF240305S1

Surrogates: Yes, QC Types: BLK/LCS/LCSD/MS/MSD

Sample ID	Analysis	Method	Run Date/Time	Batch ID
S59163.02	34 PFAs	ASTM D7968-17M	03/06/24 00:33	AK240305

QC Report - Internal Standards per Lab Sample

Lab Sample ID: S59163.01

Sample Tag: Equipment Blank

Collected Date/Time: 02/27/2024 13:20

Matrix: Water

COC Reference: 159864

Organics - Volatiles, Analysis: 34 PFAs

Run in Batch: AK240229, Run Date: 02/29/2024 23:36, Matrix: WW, Dilution: 1.76

Internal Standard	Flags	%Rec	LCL	UCL
M2-4:2FTSA		**104.0**	50.0	150.0
M2-6:2FTSA		**105.3**	50.0	150.0
M2-8:2FTSA		**101.0**	50.0	150.0
M2PFTeDA		**85.7**	12.0	218.0
M3PFBS		**100.9**	50.0	150.0
M3PFHxS		**116.4**	50.0	150.0
M4PFHpA		**101.9**	50.0	150.0
M5PFHxA		**92.2**	50.0	150.0
M5PFPeA		**99.8**	50.0	150.0
M6PFDA		**101.2**	50.0	150.0
M7PFUnDA		**95.5**	50.0	150.0
M8FOSA		**91.7**	50.0	150.0
M8PFOA		**104.7**	50.0	150.0
M8PFOS		**101.0**	50.0	150.0
M9-PFNA		**93.3**	50.0	150.0
MPFBA		**100.9**	50.0	150.0
MPFDoDA		**88.9**	50.0	150.0
d3N-MeFOSAA		**82.0**	50.0	150.0
d5EtFOSAA		**80.9**	50.0	150.0
MHFPO-DA		**114.9**	50.0	150.0
d-N-EtFOSA-M		**78.6**	50.0	150.0
d-N-MeFOSA-M		**87.8**	50.0	150.0
d7-N-MeFOSE-M		**99.5**	50.0	150.0
d9-N-EtFOSE-M		**83.0**	50.0	150.0

QC Report - Internal Standards per Lab Sample

Lab Sample ID: S59163.02

Sample Tag: Biosolids - DIG.4

Collected Date/Time: 02/27/2024 13:20

Matrix: Sludge

COC Reference: 159864

Organics - Volatiles, Analysis: 34 PFAs

Run in Batch: AK240305, Run Date: 03/06/2024 00:33, Matrix: SO, Dilution: 170

Internal Standard	Flags	%Rec	LCL	UCL
M2-4:2FTSA	*	**296.1**	50.0	150.0
M2-6:2FTSA	*	**229.5**	50.0	150.0
M2-8:2FTSA	*	**364.1**	50.0	150.0
M2PFTeDA		**54.9**	12.0	218.0
M3PFBS		**82.1**	50.0	150.0
M3PFHxS		**83.8**	50.0	150.0
M4PFHpA		**90.3**	50.0	150.0
M5PFHxA		**82.9**	50.0	150.0
M5PFPeA		**89.9**	50.0	150.0
M6PFDA		**85.3**	50.0	150.0
M7PFUnDA		**57.4**	50.0	150.0
M8FOSA		**109.1**	50.0	150.0
M8PFOA		**86.8**	50.0	150.0
M8PFOS		**88.2**	50.0	150.0
M9-PFNA		**95.2**	50.0	150.0
MPFBA		**94.8**	50.0	150.0
MPFDoDA	*	**39.1**	50.0	150.0
d3N-MeFOSAA		**148.4**	50.0	150.0
d5EtFOSAA	*	**157.6**	50.0	150.0
MHFPO-DA		**82.8**	50.0	150.0
d-N-EtFOSA-M		**63.8**	50.0	150.0
d-N-MeFOSA-M		**65.1**	50.0	150.0
d7-N-MeFOSE-M	*	**44.6**	50.0	150.0
d9-N-EtFOSE-M	*	**35.0**	50.0	150.0

Organics - Volatiles, Prep Batch ID: PF240229W1

QC Types: BLK/LCS/LCSD/MS/DUP

Blank (BLK)

Lab Sample ID: AK240229.BLK240229

Run in Batch: AK240229, Run Date: 02/29/2024 15:55, Prep Date: 02/29/2024, Matrix: WW, Dilution: 1

Internal Standard	Flags	%Rec	LCL	UCL
M2-4:2FTSA		**117.3**	50.0	150.0
M2-6:2FTSA		**127.0**	50.0	150.0
M2-8:2FTSA		**103.3**	50.0	150.0
M2PFTeDA		**93.1**	12.0	218.0
M3PFBS		**101.4**	50.0	150.0
M3PFHxS		**122.5**	50.0	150.0
M4PFHpA		**107.1**	50.0	150.0
M5PFHxA		**92.2**	50.0	150.0
M5PFPeA		**105.2**	50.0	150.0
M6PFDA		**102.9**	50.0	150.0
M7PFUnDA		**96.9**	50.0	150.0
M8FOSA		**103.7**	50.0	150.0
M8PFOA		**113.3**	50.0	150.0
M8PFOS		**116.5**	50.0	150.0
M9-PFNA		**114.7**	50.0	150.0
MPFBA		**107.9**	50.0	150.0
MPFDoDA		**97.5**	50.0	150.0
d3N-MeFOSAA		**112.6**	50.0	150.0
d5EtFOSAA		**87.5**	50.0	150.0
MHFPO-DA		**97.0**	50.0	150.0
d-N-EtFOSA-M		**91.8**	50.0	150.0
d-N-MeFOSA-M		**99.6**	50.0	150.0
d7-N-MeFOSE-M		**108.3**	50.0	150.0
d9-N-EtFOSE-M		**91.5**	50.0	150.0

QC Report - Internal Standards per QC Sample

Laboratory Control Sample (LCS)

Lab Sample ID: AK240229.LCS240229

Run in Batch: AK240229, Run Date: 02/29/2024 14:54, Prep Date: 02/29/2024, Matrix: WW, Dilution: 1

Internal Standard	Flags	%Rec	LCL	UCL
M2-4:2FTSA		**117.8**	50.0	150.0
M2-6:2FTSA		**114.2**	50.0	150.0
M2-8:2FTSA		**89.9**	50.0	150.0
M2PFTeDA		**93.7**	12.0	218.0
M3PFBS		**104.6**	50.0	150.0
M3PFHxS		**118.8**	50.0	150.0
M4PFHpA		**110.6**	50.0	150.0
M5PFHxA		**93.4**	50.0	150.0
M5PFPeA		**100.6**	50.0	150.0
M6PFDA		**93.6**	50.0	150.0
M7PFUnDA		**105.8**	50.0	150.0
M8FOSA		**99.7**	50.0	150.0
M8PFOA		**108.7**	50.0	150.0
M8PFOS		**121.0**	50.0	150.0
M9-PFNA		**100.9**	50.0	150.0
MPFBA		**106.1**	50.0	150.0
MPFDoDA		**97.4**	50.0	150.0
d3N-MeFOSAA		**107.0**	50.0	150.0
d5EtFOSAA		**106.3**	50.0	150.0
MHFPO-DA		**105.9**	50.0	150.0
d-N-EtFOSA-M		**89.4**	50.0	150.0
d-N-MeFOSA-M		**99.3**	50.0	150.0
d7-N-MeFOSE-M		**101.8**	50.0	150.0
d9-N-EtFOSE-M		**98.9**	50.0	150.0

Laboratory Control Sample Duplicate (LCSD)

Lab Sample ID: AK240229.LCSD240229, Parent Sample ID: AK240229.LCS240229

Run in Batch: AK240229, Run Date: 02/29/2024 15:35, Prep Date: 02/29/2024, Matrix: WW, Dilution: 1

Internal Standard	Flags	%Rec	LCL	UCL
M2-4:2FTSA		**105.2**	50.0	150.0
M2-6:2FTSA		**93.5**	50.0	150.0
M2-8:2FTSA		**100.5**	50.0	150.0
M2PFTeDA		**83.1**	12.0	218.0
M3PFBS		**97.4**	50.0	150.0
M3PFHxS		**108.2**	50.0	150.0
M4PFHpA		**99.5**	50.0	150.0
M5PFHxA		**88.7**	50.0	150.0
M5PFPeA		**98.4**	50.0	150.0
M6PFDA		**91.2**	50.0	150.0
M7PFUnDA		**106.9**	50.0	150.0
M8FOSA		**96.5**	50.0	150.0
M8PFOA		**98.8**	50.0	150.0
M8PFOS		**102.7**	50.0	150.0
M9-PFNA		**93.8**	50.0	150.0
MPFBA		**101.1**	50.0	150.0
MPFDoDA		**93.8**	50.0	150.0
d3N-MeFOSAA		**93.6**	50.0	150.0
d5EtFOSAA		**90.2**	50.0	150.0
MHFPO-DA		**114.4**	50.0	150.0
d-N-EtFOSA-M		**93.4**	50.0	150.0
d-N-MeFOSA-M		**94.4**	50.0	150.0
d7-N-MeFOSE-M		**98.3**	50.0	150.0
d9-N-EtFOSE-M		**92.6**	50.0	150.0

Matrix Spike (MS)

Lab Sample ID: AK240229.5906204M, Parent Sample ID: S59062.04

Run in Batch: AK240229, Run Date: 02/29/2024 19:16, Prep Date: 02/29/2024, Matrix: WW, Dilution: 2.1

Internal Standard	Flags	%Rec	LCL	UCL
M2-4:2FTSA		**114.4**	50.0	150.0
M2-6:2FTSA		**114.6**	50.0	150.0
M2-8:2FTSA		**92.8**	50.0	150.0
M2PFTeDA		**93.9**	12.0	218.0
M3PFBS		**114.9**	50.0	150.0
M3PFHxS		**128.4**	50.0	150.0
M4PFHpA		**99.7**	50.0	150.0
M5PFHxA		**100.6**	50.0	150.0
M5PFPeA		**109.2**	50.0	150.0
M6PFDA		**105.4**	50.0	150.0
M7PFUnDA		**105.5**	50.0	150.0
M8FOSA		**113.5**	50.0	150.0
M8PFOA		**109.9**	50.0	150.0
M8PFOS		**112.9**	50.0	150.0
M9-PFNA		**105.9**	50.0	150.0
MPFBA		**108.4**	50.0	150.0
MPFDoDA		**104.6**	50.0	150.0
d3N-MeFOSAA		**107.3**	50.0	150.0
d5EtFOSAA		**102.9**	50.0	150.0
MHFPO-DA		**101.8**	50.0	150.0
d-N-EtFOSA-M		**97.6**	50.0	150.0
d-N-MeFOSA-M		**103.0**	50.0	150.0
d7-N-MeFOSE-M		**104.2**	50.0	150.0
d9-N-EtFOSE-M		**100.0**	50.0	150.0

QC Report - Internal Standards per QC Sample

Duplicate (DUP)

Lab Sample ID: AK240229.5906205D, Parent Sample ID: S59062.05

Run in Batch: AK240229, Run Date: 02/29/2024 19:56, Prep Date: 02/29/2024, Matrix: WW, Dilution: 1.99

Internal Standard	Flags	%Rec	LCL	UCL
M2-4:2FTSA		**103.4**	50.0	150.0
M2-6:2FTSA		**119.5**	50.0	150.0
M2-8:2FTSA		**102.5**	50.0	150.0
M2PFTeDA		**102.5**	12.0	218.0
M3PFBS		**103.8**	50.0	150.0
M3PFHxS		**113.4**	50.0	150.0
M4PFHpA		**110.9**	50.0	150.0
M5PFHxA		**105.9**	50.0	150.0
M5PFPeA		**108.6**	50.0	150.0
M6PFDA		**104.8**	50.0	150.0
M7PFUnDA		**95.9**	50.0	150.0
M8FOSA		**103.6**	50.0	150.0
M8PFOA		**105.1**	50.0	150.0
M8PFOS		**117.5**	50.0	150.0
M9-PFNA		**108.4**	50.0	150.0
MPFBA		**109.0**	50.0	150.0
MPFDoDA		**101.6**	50.0	150.0
d3N-MeFOSAA		**118.8**	50.0	150.0
d5EtFOSAA		**93.7**	50.0	150.0
MHFPO-DA		**105.9**	50.0	150.0
d-N-EtFOSA-M		**103.0**	50.0	150.0
d-N-MeFOSA-M		**105.2**	50.0	150.0
d7-N-MeFOSE-M		**105.3**	50.0	150.0
d9-N-EtFOSE-M		**96.8**	50.0	150.0

QC Report - Internal Standards per QC Sample

Organics - Volatiles, Prep Batch ID: PF240305S1

QC Types: BLK/LCS/LCSD/MS/MSD

Blank (BLK)

Lab Sample ID: AK240305.BLK240305S

Run in Batch: AK240305, Run Date: 03/05/2024 18:33, Prep Date: 03/05/2024, Matrix: SO, Dilution: 1

Internal Standard	Flags	%Rec	LCL	UCL
M2-4:2FTSA		**123.6**	50.0	150.0
M2-6:2FTSA		**108.7**	50.0	150.0
M2-8:2FTSA		**120.7**	50.0	150.0
M2PFTeDA		**91.3**	12.0	218.0
M3PFBS		**110.4**	50.0	150.0
M3PFHxS		**104.4**	50.0	150.0
M4PFHpA		**105.7**	50.0	150.0
M5PFHxA		**114.5**	50.0	150.0
M5PFPeA		**110.5**	50.0	150.0
M6PFDA		**107.1**	50.0	150.0
M7PFUnDA		**88.4**	50.0	150.0
M8FOSA		**101.8**	50.0	150.0
M8PFOA		**112.2**	50.0	150.0
M8PFOS		**116.4**	50.0	150.0
M9-PFNA		**115.5**	50.0	150.0
MPFBA		**110.5**	50.0	150.0
MPFDoDA		**87.3**	50.0	150.0
d3N-MeFOSAA		**107.0**	50.0	150.0
d5EtFOSAA		**113.8**	50.0	150.0
MHFPO-DA		**105.8**	50.0	150.0
d-N-EtFOSA-M		**92.9**	50.0	150.0
d-N-MeFOSA-M		**92.4**	50.0	150.0
d7-N-MeFOSE-M		**76.1**	50.0	150.0
d9-N-EtFOSE-M		**80.6**	50.0	150.0

QC Report - Internal Standards per QC Sample

Laboratory Control Sample (LCS)

Lab Sample ID: AK240305.LCS240305

Run in Batch: AK240305, Run Date: 03/05/2024 17:53, Prep Date: 03/05/2024, Matrix: SO, Dilution: 1

Internal Standard	Flags	%Rec	LCL	UCL
M2-4:2FTSA		**120.7**	50.0	150.0
M2-6:2FTSA		**111.7**	50.0	150.0
M2-8:2FTSA		**121.2**	50.0	150.0
M2PFTeDA		**96.5**	12.0	218.0
M3PFBS		**108.8**	50.0	150.0
M3PFHxS		**94.4**	50.0	150.0
M4PFHpA		**93.2**	50.0	150.0
M5PFHxA		**98.9**	50.0	150.0
M5PFPeA		**100.9**	50.0	150.0
M6PFDA		**90.3**	50.0	150.0
M7PFUnDA		**91.1**	50.0	150.0
M8FOSA		**99.1**	50.0	150.0
M8PFOA		**106.8**	50.0	150.0
M8PFOS		**114.9**	50.0	150.0
M9-PFNA		**101.1**	50.0	150.0
MPFBA		**102.5**	50.0	150.0
MPFDoDA		**93.2**	50.0	150.0
d3N-MeFOSAA		**102.5**	50.0	150.0
d5EtFOSAA		**106.5**	50.0	150.0
MHFPO-DA		**107.3**	50.0	150.0
d-N-EtFOSA-M		**93.5**	50.0	150.0
d-N-MeFOSA-M		**85.2**	50.0	150.0
d7-N-MeFOSE-M		**80.0**	50.0	150.0
d9-N-EtFOSE-M		**81.1**	50.0	150.0

QC Report - Internal Standards per QC Sample

Laboratory Control Sample Duplicate (LCSD)

Lab Sample ID: AK240305.LCSD240305, Parent Sample ID: AK240305.LCS240305

Run in Batch: AK240305, Run Date: 03/05/2024 18:13, Prep Date: 03/05/2024, Matrix: SO, Dilution: 1

Internal Standard	Flags	%Rec	LCL	UCL
M2-4:2FTSA		**116.5**	50.0	150.0
M2-6:2FTSA		**100.6**	50.0	150.0
M2-8:2FTSA		**107.6**	50.0	150.0
M2PFTeDA		**91.2**	12.0	218.0
M3PFBS		**104.6**	50.0	150.0
M3PFHxS		**103.9**	50.0	150.0
M4PFHpA		**102.0**	50.0	150.0
M5PFHxA		**109.6**	50.0	150.0
M5PFPeA		**101.0**	50.0	150.0
M6PFDA		**93.2**	50.0	150.0
M7PFUnDA		**89.9**	50.0	150.0
M8FOSA		**100.6**	50.0	150.0
M8PFOA		**103.2**	50.0	150.0
M8PFOS		**101.2**	50.0	150.0
M9-PFNA		**96.4**	50.0	150.0
MPFBA		**103.7**	50.0	150.0
MPFDoDA		**86.2**	50.0	150.0
d3N-MeFOSAA		**96.0**	50.0	150.0
d5EtFOSAA		**100.2**	50.0	150.0
MHFPO-DA		**82.3**	50.0	150.0
d-N-EtFOSA-M		**92.0**	50.0	150.0
d-N-MeFOSA-M		**95.0**	50.0	150.0
d7-N-MeFOSE-M		**69.5**	50.0	150.0
d9-N-EtFOSE-M		**76.2**	50.0	150.0

Matrix Spike (MS)

Lab Sample ID: AK240305.5878403REXM, Parent Sample ID: S58784.03

Run in Batch: AK240305, Run Date: 03/05/2024 19:33, Prep Date: 03/05/2024, Matrix: SO, Dilution: 4.75

Internal Standard	Flags	%Rec	LCL	UCL
M2-4:2FTSA	*	**338.2**	50.0	150.0
M2-6:2FTSA	*	**345.5**	50.0	150.0
M2-8:2FTSA	*	**477.4**	50.0	150.0
M2PFTeDA		**16.8**	12.0	218.0
M3PFBS		**92.8**	50.0	150.0
M3PFHxS		**91.1**	50.0	150.0
M4PFHpA		**79.3**	50.0	150.0
M5PFHxA		**87.0**	50.0	150.0
M5PFPeA		**89.8**	50.0	150.0
M6PFDA		**82.1**	50.0	150.0
M7PFUnDA		**93.3**	50.0	150.0
M8FOSA		**116.8**	50.0	150.0
M8PFOA		**91.9**	50.0	150.0
M8PFOS		**100.5**	50.0	150.0
M9-PFNA		**91.5**	50.0	150.0
MPFBA		**87.0**	50.0	150.0
MPFDoDA		**65.2**	50.0	150.0
d3N-MeFOSAA	*	**150.2**	50.0	150.0
d5EtFOSAA	*	**192.8**	50.0	150.0
MHFPO-DA		**67.3**	50.0	150.0
d-N-EtFOSA-M		**64.0**	50.0	150.0
d-N-MeFOSA-M		**75.9**	50.0	150.0
d7-N-MeFOSE-M	*	**49.0**	50.0	150.0
d9-N-EtFOSE-M	*	**47.6**	50.0	150.0

Matrix Spike Duplicate (MSD)

Lab Sample ID: AK240305.5878403REXMD, Parent Sample ID: AK240305.5878403REXM

Run in Batch: AK240305, Run Date: 03/05/2024 19:53, Prep Date: 03/05/2024, Matrix: SO, Dilution: 4.89

Internal Standard	Flags	**%Rec**	LCL	UCL
M2-4:2FTSA	*	**315.3**	50.0	150.0
M2-6:2FTSA	*	**391.6**	50.0	150.0
M2-8:2FTSA	*	**393.3**	50.0	150.0
M2PFTeDA		**37.2**	12.0	218.0
M3PFBS		**90.4**	50.0	150.0
M3PFHxS		**99.9**	50.0	150.0
M4PFHpA		**85.7**	50.0	150.0
M5PFHxA		**84.0**	50.0	150.0
M5PFPeA		**87.3**	50.0	150.0
M6PFDA		**85.9**	50.0	150.0
M7PFUnDA		**95.9**	50.0	150.0
M8FOSA		**113.8**	50.0	150.0
M8PFOA		**89.4**	50.0	150.0
M8PFOS		**105.1**	50.0	150.0
M9-PFNA		**88.5**	50.0	150.0
MPFBA		**89.4**	50.0	150.0
MPFDoDA		**76.0**	50.0	150.0
d3N-MeFOSAA	*	**161.9**	50.0	150.0
d5EtFOSAA	*	**173.8**	50.0	150.0
MHFPO-DA		**74.5**	50.0	150.0
d-N-EtFOSA-M		**67.5**	50.0	150.0
d-N-MeFOSA-M		**76.8**	50.0	150.0
d7-N-MeFOSE-M		**51.1**	50.0	150.0
d9-N-EtFOSE-M	*	**42.3**	50.0	150.0

Inorganics, Prep Batch ID: TS240228A

Surrogates: No, QC Types: BLK/LCS/DUP

Blank (BLK)

Lab Sample ID: TS240228A.LRB1

Run in Batch: TS240228A, Run Date: 02/28/2024 12:39, Prep Date: 02/28/2024, Matrix: Liquid, Dilution: 1

Analyte	Flags	Conc	RDL	Units
Total Solids		ND	1	%

Laboratory Control Sample (LCS)

Lab Sample ID: TS240228A.LCS1

Run in Batch: TS240228A, Run Date: 02/28/2024 12:39, Prep Date: 02/28/2024, Matrix: Liquid, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL
Total Solids		100	90	110

Duplicate (DUP)

Lab Sample ID: TS240228A.DP1, Parent Sample ID: S59140.01

Run in Batch: TS240228A, Run Date: 02/28/2024 12:39, Prep Date: 02/28/2024, Matrix: Soil, Dilution: 1

Analyte	Flags	RPD	RPD CL
Total Solids		1	5

Duplicate (DUP)

Lab Sample ID: TS240228A.DP2, Parent Sample ID: S59153.02

Run in Batch: TS240228A, Run Date: 02/28/2024 12:39, Prep Date: 02/28/2024, Matrix: Soil, Dilution: 1

Analyte	Flags	RPD	RPD CL
Total Solids		0	5

Organics - Volatiles, Prep Batch ID: PF240229W1
Surrogates: Yes, QC Types: BLK/LCS/LCSD/MS/DUP

Blank (BLK)
Lab Sample ID: AK240229.BLK240229
Run in Batch: AK240229, Run Date: 02/29/2024 15:55, Prep Date: 02/29/2024, Matrix: WW, Dilution: 1

Analyte	Flags	Conc	RDL	Units
PFBA		ND	10	ng/l
PFMPA		ND	2	ng/l
FPrPA (3:3 FTCA)		ND	10	ng/l
PFPrS		ND	2	ng/l
PFPeA		ND	4	ng/l
PFMBA		ND	2	ng/l
4:2 FTSA		ND	2	ng/l
NFDHA		ND	2	ng/l
PFHxA		ND	2	ng/l
PFBS		ND	2	ng/l
HFPO-DA		ND	10	ng/l
FPePA (5:3 FTCA)		ND	10	ng/l
PFEESA		ND	2	ng/l
PFHpA		ND	2	ng/l
ADONA		ND	2	ng/l
PFPeS		ND	2	ng/l
6:2 FTSA		ND	2	ng/l
PFBSA		ND	2	ng/l
PFOA		ND	2	ng/l
PFHxS-BR		ND	2	ng/l
PFHxS		ND	2	ng/l
PFHxS-LN		ND	2	ng/l
FHpPA (7:3 FTCA)		ND	10	ng/l
PFNA		ND	2	ng/l
8:2 FTSA		ND	2	ng/l
PFECHS		ND	2	ng/l
PFHpS		ND	2	ng/l
N-MeFOSAA		ND	2	ng/l
PFDA		ND	2	ng/l
PFOS-BR		ND	2	ng/l
EtFOSAA		ND	4	ng/l
PFOS		ND	2	ng/l
PFOS-LN		ND	2	ng/l
PFHxSA		ND	2	ng/l
PFUnDA		ND	2	ng/l
9CL-PF3ONS		ND	2	ng/l
PFNS		ND	2	ng/l
PFDoDA		ND	2	ng/l
PFDS		ND	2	ng/l
PFTrDA		ND	2	ng/l
FOSA		ND	2	ng/l
11CL-PF3OUdS		ND	2	ng/l
PFTeDA		ND	4	ng/l
PFDOS		ND	6	ng/l
NMeFOSE		ND	4	ng/l
NMeFOSAM		ND	2	ng/l

Organics - Volatiles, Prep Batch ID: PF240229W1 (continued)
Surrogates: Yes, QC Types: BLK/LCS/LCSD/MS/DUP

Blank (BLK) (continued)
Lab Sample ID: AK240229.BLK240229
Run in Batch: AK240229, Run Date: 02/29/2024 15:55, Prep Date: 02/29/2024, Matrix: WW, Dilution: 1

Analyte	Flags	Conc	RDL	Units
NEtFOSE		ND	4	ng/l
NEtFOSAM		ND	2	ng/l

Laboratory Control Sample (LCS)
Lab Sample ID: AK240229.LCS240229
Run in Batch: AK240229, Run Date: 02/29/2024 14:54, Prep Date: 02/29/2024, Matrix: WW, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL
PFBA		99.2	70.0	130.0
PFMPA		74.4	70.0	130.0
FPrPA (3:3 FTCA)		100.8	70.0	130.0
PFPrS		96.2	70.0	130.0
PFPeA		95.6	70.0	130.0
PFMBA		80.6	70.0	130.0
4:2 FTSA		87.8	70.0	130.0
NFDHA		76.0	70.0	130.0
PFHxA		92.6	70.0	130.0
PFBS		89.4	70.0	130.0
HFPO-DA		96.6	70.0	130.0
FPePA (5:3 FTCA)		93.8	70.0	130.0
PFEESA		83.6	70.0	130.0
PFHpA		93.0	70.0	130.0
ADONA		88.8	70.0	130.0
PFPeS		86.8	70.0	130.0
6:2 FTSA		107.0	70.0	130.0
PFBSA		95.4	70.0	130.0
PFOA		82.0	70.0	130.0
PFHxS		89.4	70.0	130.0
FHpPA (7:3 FTCA)		81.6	70.0	130.0
PFNA		105.4	70.0	130.0
8:2 FTSA		110.8	70.0	130.0
PFECHS		102.8	70.0	130.0
PFHpS		84.2	70.0	130.0
N-MeFOSAA		109.6	70.0	130.0
PFDA		105.2	70.0	130.0
EtFOSAA		85.8	70.0	130.0
PFOS		95.4	70.0	130.0
PFHxSA		87.2	70.0	130.0
PFUnDA		97.2	70.0	130.0
9CL-PF3ONS		87.8	70.0	130.0
PFNS		95.4	70.0	130.0
PFDoDA		93.2	70.0	130.0
PFDS		102.6	70.0	130.0
PFTrDA		103.2	70.0	130.0
FOSA		94.4	70.0	130.0
11CL-PF3OUdS		101.6	70.0	130.0
PFTeDA		87.8	70.0	130.0

Organics - Volatiles, Prep Batch ID: PF240229W1 (continued)
Surrogates: Yes, QC Types: BLK/LCS/LCSD/MS/DUP

Laboratory Control Sample (LCS) (continued)
Lab Sample ID: AK240229.LCS240229
Run in Batch: AK240229, Run Date: 02/29/2024 14:54, Prep Date: 02/29/2024, Matrix: WW, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL
PFDOS		99.6	70.0	130.0
NMeFOSE		103.4	70.0	130.0
NMeFOSAM		96.0	70.0	130.0
NEtFOSE		99.8	70.0	130.0
NEtFOSAM		104.2	70.0	130.0

Laboratory Control Sample Duplicate (LCSD)
Lab Sample ID: AK240229.LCSD240229, Parent Sample ID: AK240229.LCS240229
Run in Batch: AK240229, Run Date: 02/29/2024 15:35, Prep Date: 02/29/2024, Matrix: WW, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL	RPD	RPD CL
PFBA		105.4	70.0	130.0	6.1	30.0
PFMPA		85.8	70.0	130.0	14.2	30.0
FPrPA (3:3 FTCA)		100.4	70.0	130.0	0.4	30.0
PFPrS		96.2	70.0	130.0	0.0	30.0
PFPeA		97.6	70.0	130.0	2.1	30.0
PFMBA		97.6	70.0	130.0	19.1	30.0
4:2 FTSA		106.0	70.0	130.0	18.8	30.0
NFDHA		99.0	70.0	130.0	26.3	30.0
PFHxA		98.2	70.0	130.0	5.9	30.0
PFBS		98.6	70.0	130.0	9.8	30.0
HFPO-DA		90.8	70.0	130.0	6.2	30.0
FPePA (5:3 FTCA)		92.4	70.0	130.0	1.5	30.0
PFEESA		96.0	70.0	130.0	13.8	30.0
PFHpA		105.2	70.0	130.0	12.3	30.0
ADONA		103.6	70.0	130.0	15.4	30.0
PFPeS		97.8	70.0	130.0	11.9	30.0
6:2 FTSA		121.0	70.0	130.0	12.3	30.0
PFBSA		100.8	70.0	130.0	5.5	30.0
PFOA		90.4	70.0	130.0	9.7	30.0
PFHxS		99.4	70.0	130.0	10.6	30.0
FHpPA (7:3 FTCA)		93.4	70.0	130.0	13.5	30.0
PFNA		109.8	70.0	130.0	4.1	30.0
8:2 FTSA		95.6	70.0	130.0	14.7	30.0
PFECHS		122.8	70.0	130.0	17.7	30.0
PFHpS		105.0	70.0	130.0	22.0	30.0
N-MeFOSAA		115.2	70.0	130.0	5.0	30.0
PFDA		105.8	70.0	130.0	0.6	30.0
EtFOSAA		108.6	70.0	130.0	23.5	30.0
PFOS		111.2	70.0	130.0	15.3	30.0
PFHxSA		99.4	70.0	130.0	13.1	30.0
PFUnDA		84.6	70.0	130.0	13.9	30.0
9CL-PF3ONS		104.4	70.0	130.0	17.3	30.0
PFNS		104.4	70.0	130.0	9.0	30.0
PFDoDA		99.6	70.0	130.0	6.6	30.0
PFDS		109.4	70.0	130.0	6.4	30.0
PFTrDA		97.2	70.0	130.0	6.0	30.0

Organics - Volatiles, Prep Batch ID: PF240229W1 (continued)
Surrogates: Yes, QC Types: BLK/LCS/LCSD/MS/DUP

Laboratory Control Sample Duplicate (LCSD) (continued)
Lab Sample ID: AK240229.LCSD240229, Parent Sample ID: AK240229.LCS240229
Run in Batch: AK240229, Run Date: 02/29/2024 15:35, Prep Date: 02/29/2024, Matrix: WW, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL	RPD	RPD CL
FOSA		93.8	70.0	130.0	0.6	30.0
11CL-PF3OUdS		113.8	70.0	130.0	11.3	30.0
PFTeDA		99.2	70.0	130.0	12.2	30.0
PFDOS		121.0	70.0	130.0	19.4	30.0
NMeFOSE		114.6	70.0	130.0	10.3	30.0
NMeFOSAM		106.0	70.0	130.0	9.9	30.0
NEtFOSE		101.6	70.0	130.0	1.8	30.0
NEtFOSAM		101.4	70.0	130.0	2.7	30.0

Matrix Spike (MS)
Lab Sample ID: AK240229.5906204M, Parent Sample ID: S59062.04
Run in Batch: AK240229, Run Date: 02/29/2024 19:16, Prep Date: 02/29/2024, Matrix: WW, Dilution: 2.1

Analyte	Flags	% Rec	LCL	UCL
PFBA		104.8	70.0	130.0
PFPeA		89.5	70.0	130.0
4:2 FTSA		86.7	70.0	130.0
PFHxA		80.0	70.0	130.0
PFBS		82.9	70.0	130.0
PFHpA		95.2	70.0	130.0
PFPeS		85.7	70.0	130.0
6:2 FTSA		95.2	70.0	130.0
PFOA		79.0	70.0	130.0
PFHxS		77.1	70.0	130.0
PFNA		95.2	70.0	130.0
8:2 FTSA		104.8	70.0	130.0
PFHpS		93.3	70.0	130.0
PFDA		92.4	70.0	130.0
N-MeFOSAA		104.8	70.0	130.0
EtFOSAA		95.2	70.0	130.0
PFOS		104.8	70.0	130.0
PFUnDA		104.8	70.0	130.0
PFNS		104.8	70.0	130.0
PFDoDA		87.6	70.0	130.0
PFDS		104.8	70.0	130.0
PFTrDA		92.4	70.0	130.0
FOSA		82.9	70.0	130.0
PFTeDA		88.6	70.0	130.0
11CL-PF3OUdS		104.8	70.0	130.0
9CL-PF3ONS		104.8	70.0	130.0
ADONA		83.8	70.0	130.0
HFPO-DA		114.3	70.0	130.0
FHpPA (7:3 FTCA)		86.7	70.0	130.0
FPePA (5:3 FTCA)		88.6	70.0	130.0
FPrPA (3:3 FTCA)		95.2	70.0	130.0
NFDHA		87.6	70.0	130.0
PFBSA		92.4	70.0	130.0

Organics - Volatiles, Prep Batch ID: PF240229W1 (continued)

Surrogates: Yes, QC Types: BLK/LCS/LCSD/MS/DUP

Matrix Spike (MS) (continued)

Lab Sample ID: AK240229.5906204M, Parent Sample ID: S59062.04

Run in Batch: AK240229, Run Date: 02/29/2024 19:16, Prep Date: 02/29/2024, Matrix: WW, Dilution: 2.1

Analyte	Flags	% Rec	LCL	UCL
PFECHS		104.8	70.0	130.0
PFEESA		92.4	70.0	130.0
PFHxSA		85.7	70.0	130.0
PFMBA		85.7	70.0	130.0
PFMPA		82.9	70.0	130.0
PFPrS		89.5	70.0	130.0

Duplicate (DUP)

Lab Sample ID: AK240229.5906205D, Parent Sample ID: S59062.05

Run in Batch: AK240229, Run Date: 02/29/2024 19:56, Prep Date: 02/29/2024, Matrix: WW, Dilution: 1.99

Analyte	Flags	RPD	RPD CL
PFBA		NC	30.0
PFPeA		NC	30.0
4:2 FTSA		NC	30.0
PFHxA		NC	30.0
PFBS		NC	30.0
PFHpA		NC	30.0
PFPeS		NC	30.0
6:2 FTSA		NC	30.0
PFOA		NC	30.0
PFHxS		NC	30.0
PFHxS-LN		NC	30.0
PFHxS-BR		NC	30.0
PFNA		NC	30.0
8:2 FTSA		NC	30.0
PFHpS		NC	30.0
PFDA		NC	30.0
N-MeFOSAA		NC	30.0
EtFOSAA		NC	30.0
PFOS		NC	30.0
PFOS-LN		NC	30.0
PFOS-BR		NC	30.0
PFUnDA		NC	30.0
PFNS		NC	30.0
PFDoDA		NC	30.0
PFDS		NC	30.0
PFTrDA		NC	30.0
FOSA		NC	30.0
PFTeDA		NC	30.0
11CL-PF3OUdS		NC	30.0
9CL-PF3ONS		NC	30.0
ADONA		NC	30.0
HFPO-DA		NC	30.0
FHpPA (7:3 FTCA)		NC	30.0
FPePA (5:3 FTCA)		NC	30.0
FPrPA (3:3 FTCA)		NC	30.0

Organics - Volatiles, Prep Batch ID: PF240229W1 (continued)

Surrogates: Yes, QC Types: BLK/LCS/LCSD/MS/DUP

Duplicate (DUP) (continued)

Lab Sample ID: AK240229.5906205D, Parent Sample ID: S59062.05

Run in Batch: AK240229, Run Date: 02/29/2024 19:56, Prep Date: 02/29/2024, Matrix: WW, Dilution: 1.99

Analyte	Flags	RPD	RPD CL
NFDHA		NC	30.0
PFBSA		NC	30.0
PFECHS		NC	30.0
PFEESA		NC	30.0
PFHxSA		NC	30.0
PFMBA		NC	30.0
PFMPA		NC	30.0
PFPrS		NC	30.0

Organics - Volatiles, Prep Batch ID: PF240305S1
Surrogates: Yes, QC Types: BLK/LCS/LCSD/MS/MSD

Blank (BLK)
Lab Sample ID: AK240305.BLK240305S
Run in Batch: AK240305, Run Date: 03/05/2024 18:33, Prep Date: 03/05/2024, Matrix: SO, Dilution: 1

Analyte	Flags	Conc	RDL	Units
PFBA		ND	20	ng/kg
PFMPA		ND	10	ng/kg
FPrPA (3:3 FTCA)		ND	10	ng/kg
PFPrS		ND	10	ng/kg
PFPeA		ND	10	ng/kg
PFMBA		ND	10	ng/kg
4:2 FTSA		ND	10	ng/kg
NFDHA		ND	10	ng/kg
PFHxA		ND	10	ng/kg
PFBS		ND	10	ng/kg
HFPO-DA		ND	10	ng/kg
FPePA (5:3 FTCA)		ND	10	ng/kg
PFEESA		ND	10	ng/kg
PFHpA		ND	10	ng/kg
PFPeS		ND	10	ng/kg
ADONA		ND	10	ng/kg
6:2 FTSA		ND	10	ng/kg
PFBSA		ND	10	ng/kg
PFOA		ND	10	ng/kg
PFHxS-BR		ND	10	ng/kg
PFHxS		ND	10	ng/kg
PFHxS-LN		ND	10	ng/kg
FHpPA (7:3 FTCA)		ND	10	ng/kg
PFNA		ND	10	ng/kg
PFECHS		ND	10	ng/kg
8:2 FTSA		ND	10	ng/kg
PFHpS		ND	10	ng/kg
N-MeFOSAA		ND	10	ng/kg
PFDA		ND	10	ng/kg
PFOS-BR		ND	10	ng/kg
EtFOSAA		ND	10	ng/kg
PFOS		ND	10	ng/kg
PFOS-LN		ND	10	ng/kg
PFHxSA		ND	10	ng/kg
PFUnDA		ND	10	ng/kg
9CL-PF3ONS		ND	10	ng/kg
PFNS		ND	10	ng/kg
PFDoDA		ND	10	ng/kg
PFDS		ND	10	ng/kg
PFTrDA		ND	10	ng/kg
FOSA		ND	10	ng/kg
11CL-PF3OUdS		ND	10	ng/kg
PFTeDA		ND	10	ng/kg
PFDOS		ND	10	ng/kg
NMeFOSE		ND	10	ng/kg
NMeFOSAM		ND	10	ng/kg

Organics - Volatiles, Prep Batch ID: PF240305S1 (continued)
Surrogates: Yes, QC Types: BLK/LCS/LCSD/MS/MSD

Blank (BLK) (continued)
Lab Sample ID: AK240305.BLK240305S
Run in Batch: AK240305, Run Date: 03/05/2024 18:33, Prep Date: 03/05/2024, Matrix: SO, Dilution: 1

Analyte	Flags	Conc	RDL	Units
NEtFOSE		ND	10	ng/kg
NEtFOSAM		ND	10	ng/kg

Laboratory Control Sample (LCS)
Lab Sample ID: AK240305.LCS240305
Run in Batch: AK240305, Run Date: 03/05/2024 17:53, Prep Date: 03/05/2024, Matrix: SO, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL
PFBA		96.6	70.0	130.0
PFMPA		73.6	70.0	130.0
FPrPA (3:3 FTCA)		93.8	70.0	130.0
PFPrS		83.2	70.0	130.0
PFPeA		84.4	70.0	130.0
PFMBA		78.4	70.0	130.0
4:2 FTSA		73.8	70.0	130.0
NFDHA		77.2	70.0	130.0
PFHxA		82.6	70.0	130.0
PFBS		80.2	70.0	130.0
HFPO-DA		79.0	70.0	130.0
FPePA (5:3 FTCA)		96.6	70.0	130.0
PFEESA		80.8	70.0	130.0
PFHpA		99.2	70.0	130.0
PFPeS		88.0	70.0	130.0
ADONA		76.6	70.0	130.0
6:2 FTSA		87.0	70.0	130.0
PFBSA		89.4	70.0	130.0
PFOA		73.0	70.0	130.0
PFHxS		79.8	70.0	130.0
FHpPA (7:3 FTCA)		88.0	70.0	130.0
PFNA		89.8	70.0	130.0
PFECHS		97.2	70.0	130.0
8:2 FTSA		74.8	70.0	130.0
PFHpS		96.0	70.0	130.0
N-MeFOSAA		90.8	70.0	130.0
PFDA		90.2	70.0	130.0
EtFOSAA		94.0	70.0	130.0
PFOS		87.4	70.0	130.0
PFHxSA		87.4	70.0	130.0
PFUnDA		101.0	70.0	130.0
9CL-PF3ONS		88.6	70.0	130.0
PFNS		90.8	70.0	130.0
PFDoDA		82.8	70.0	130.0
PFDS		91.0	70.0	130.0
PFTrDA		92.0	70.0	130.0
FOSA		82.4	70.0	130.0
11CL-PF3OUdS		91.4	70.0	130.0
PFTeDA		75.8	70.0	130.0

Organics - Volatiles, Prep Batch ID: PF240305S1 (continued)

Surrogates: Yes, QC Types: BLK/LCS/LCSD/MS/MSD

Laboratory Control Sample (LCS) (continued)

Lab Sample ID: AK240305.LCS240305

Run in Batch: AK240305, Run Date: 03/05/2024 17:53, Prep Date: 03/05/2024, Matrix: SO, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL
PFDOS		95.2	70.0	130.0
NMeFOSE		99.8	70.0	130.0
NMeFOSAM		97.0	70.0	130.0
NEtFOSE		98.8	70.0	130.0
NEtFOSAM		90.8	70.0	130.0

Laboratory Control Sample Duplicate (LCSD)

Lab Sample ID: AK240305.LCSD240305, Parent Sample ID: AK240305.LCS240305

Run in Batch: AK240305, Run Date: 03/05/2024 18:13, Prep Date: 03/05/2024, Matrix: SO, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL	RPD	RPD CL
PFBA		104.0	70.0	130.0	7.4	30.0
PFMPA		91.6	70.0	130.0	21.8	30.0
FPrPA (3:3 FTCA)		115.4	70.0	130.0	20.7	30.0
PFPrS		96.6	70.0	130.0	14.9	30.0
PFPeA		98.6	70.0	130.0	15.5	30.0
PFMBA		93.4	70.0	130.0	17.5	30.0
4:2 FTSA		88.4	70.0	130.0	18.0	30.0
NFDHA		94.4	70.0	130.0	20.0	30.0
PFHxA		83.6	70.0	130.0	1.2	30.0
PFBS		96.0	70.0	130.0	17.9	30.0
HFPO-DA	*	120.2	70.0	130.0	41.4	30.0
FPePA (5:3 FTCA)		95.6	70.0	130.0	1.0	30.0
PFEESA		103.2	70.0	130.0	24.3	30.0
PFHpA		92.8	70.0	130.0	6.7	30.0
PFPeS		89.0	70.0	130.0	1.1	30.0
ADONA		97.8	70.0	130.0	24.3	30.0
6:2 FTSA		103.0	70.0	130.0	16.8	30.0
PFBSA		90.4	70.0	130.0	1.1	30.0
PFOA		86.2	70.0	130.0	16.6	30.0
PFHxS		96.4	70.0	130.0	18.8	30.0
FHpPA (7:3 FTCA)		112.2	70.0	130.0	24.2	30.0
PFNA		100.2	70.0	130.0	10.9	30.0
PFECHS		120.8	70.0	130.0	21.7	30.0
8:2 FTSA		88.0	70.0	130.0	16.2	30.0
PFHpS		95.6	70.0	130.0	0.4	30.0
N-MeFOSAA		112.4	70.0	130.0	21.3	30.0
PFDA		97.8	70.0	130.0	8.1	30.0
EtFOSAA		97.0	70.0	130.0	3.1	30.0
PFOS		102.6	70.0	130.0	16.0	30.0
PFHxSA		78.0	70.0	130.0	11.4	30.0
PFUnDA		109.0	70.0	130.0	7.6	30.0
9CL-PF3ONS		109.4	70.0	130.0	21.0	30.0
PFNS		103.8	70.0	130.0	13.4	30.0
PFDoDA		92.2	70.0	130.0	10.7	30.0
PFDS		112.2	70.0	130.0	20.9	30.0
PFTrDA		102.8	70.0	130.0	11.1	30.0

Organics - Volatiles, Prep Batch ID: PF240305S1 (continued)
Surrogates: Yes, QC Types: BLK/LCS/LCSD/MS/MSD

Laboratory Control Sample Duplicate (LCSD) (continued)
Lab Sample ID: AK240305.LCSD240305, Parent Sample ID: AK240305.LCS240305
Run in Batch: AK240305, Run Date: 03/05/2024 18:13, Prep Date: 03/05/2024, Matrix: SO, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL	RPD	RPD CL
FOSA		88.2	70.0	130.0	6.8	30.0
11CL-PF3OUdS		116.2	70.0	130.0	23.9	30.0
PFTeDA		82.4	70.0	130.0	8.3	30.0
PFDOS		116.2	70.0	130.0	19.9	30.0
NMeFOSE		118.4	70.0	130.0	17.0	30.0
NMeFOSAM		96.4	70.0	130.0	0.6	30.0
NEtFOSE		102.8	70.0	130.0	4.0	30.0
NEtFOSAM		97.6	70.0	130.0	7.2	30.0

Matrix Spike (MS)
Lab Sample ID: AK240305.5878403REXM, Parent Sample ID: S58784.03
Run in Batch: AK240305, Run Date: 03/05/2024 19:33, Prep Date: 03/05/2024, Matrix: SO, Dilution: 4.75

Analyte	Flags	% Rec	LCL	UCL
PFBA		113.4	70.0	130.0
PFPeA		92.4	70.0	130.0
4:2 FTSA		92.4	70.0	130.0
PFHxA		121.8	70.0	130.0
PFBS		84.0	70.0	130.0
PFHpA		105.0	70.0	130.0
PFPeS		79.8	70.0	130.0
6:2 FTSA		113.4	70.0	130.0
PFOA		84.0	70.0	130.0
PFHxS		100.8	70.0	130.0
PFNA		105.0	70.0	130.0
8:2 FTSA		79.8	70.0	130.0
PFHpS		109.2	70.0	130.0
PFDA		126.1	70.0	130.0
N-MeFOSAA		117.6	70.0	130.0
EtFOSAA		109.2	70.0	130.0
PFOS		79.8	70.0	130.0
PFUnDA		109.2	70.0	130.0
PFNS		100.8	70.0	130.0
PFDoDA		105.0	70.0	130.0
PFDS		105.0	70.0	130.0
PFTrDA		71.4	70.0	130.0
FOSA		92.4	70.0	130.0
PFTeDA		92.4	70.0	130.0
11CL-PF3OUdS		92.4	70.0	130.0
9CL-PF3ONS		96.6	70.0	130.0
ADONA		92.4	70.0	130.0
HFPO-DA		100.8	70.0	130.0
FHpPA (7:3 FTCA)	*	214.3	70.0	130.0
FPePA (5:3 FTCA)	*	189.1	70.0	130.0
FPrPA (3:3 FTCA)		121.8	70.0	130.0
PFBSA		75.6	70.0	130.0
PFECHS		113.4	70.0	130.0

Organics - Volatiles, Prep Batch ID: PF240305S1 (continued)

Surrogates: Yes, QC Types: BLK/LCS/LCSD/MS/MSD

Matrix Spike (MS) (continued)

Lab Sample ID: AK240305.5878403REXM, Parent Sample ID: S58784.03

Run in Batch: AK240305, Run Date: 03/05/2024 19:33, Prep Date: 03/05/2024, Matrix: SO, Dilution: 4.75

Analyte	Flags	% Rec	LCL	UCL
PFHxSA		88.2	70.0	130.0

Matrix Spike Duplicate (MSD)

Lab Sample ID: AK240305.5878403REXMD, Parent Sample ID: AK240305.5878403REXM

Run in Batch: AK240305, Run Date: 03/05/2024 19:53, Prep Date: 03/05/2024, Matrix: SO, Dilution: 4.89

Analyte	Flags	% Rec	LCL	UCL	RPD	RPD CL
PFBA		122.4	70.0	130.0	10.5	30.0
PFPeA		98.0	70.0	130.0	8.7	30.0
4:2 FTSA		93.9	70.0	130.0	4.4	30.0
PFHxA		126.5	70.0	130.0	6.7	30.0
PFBS		98.0	70.0	130.0	18.2	30.0
PFHpA		93.9	70.0	130.0	8.3	30.0
PFPeS		98.0	70.0	130.0	23.3	30.0
6:2 FTSA	*	216.3	70.0	130.0	65.0	30.0
PFOA		98.0	70.0	130.0	18.2	30.0
PFHxS		93.9	70.0	130.0	4.3	30.0
PFNA		110.2	70.0	130.0	7.7	30.0
8:2 FTSA	*	155.1	70.0	130.0	66.7	30.0
PFHpS		85.7	70.0	130.0	21.3	30.0
PFDA		106.1	70.0	130.0	14.3	30.0
N-MeFOSAA		110.2	70.0	130.0	3.6	30.0
EtFOSAA		118.4	70.0	130.0	10.9	30.0
PFOS		85.7	70.0	130.0	6.1	30.0
PFUnDA		98.0	70.0	130.0	8.0	30.0
PFNS		85.7	70.0	130.0	13.3	30.0
PFDoDA		110.2	70.0	130.0	7.7	30.0
PFDS		98.0	70.0	130.0	4.1	30.0
PFTrDA	*	98.0	70.0	130.0	34.1	30.0
FOSA		85.7	70.0	130.0	4.7	30.0
PFTeDA		93.9	70.0	130.0	4.4	30.0
11CL-PF3OUdS		89.8	70.0	130.0	0.0	30.0
9CL-PF3ONS		93.9	70.0	130.0	0.0	30.0
ADONA		81.6	70.0	130.0	9.5	30.0
HFPO-DA		98.0	70.0	130.0	0.0	30.0
FHpPA (7:3 FTCA)	*	195.9	70.0	130.0	6.1	30.0
FPePA (5:3 FTCA)	*	183.7	70.0	130.0	0.0	30.0
FPrPA (3:3 FTCA)	*	130.6	70.0	130.0	9.8	30.0
PFBSA		77.6	70.0	130.0	5.4	30.0
PFECHS		114.3	70.0	130.0	3.6	30.0
PFHxSA		89.8	70.0	130.0	4.7	30.0

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C.O.C. PAGE # 1 OF 1

159864

CHAIN OF CUSTODY RECORD

REPORT TO

CONTACT NAME: CINDY TOMASZEWSKI
COMPANY: CITY OF CADILLAC - WWTP
ADDRESS: 1121 PLETT RD
CITY: CADILLAC MI STATE: MI ZIP CODE: 49601
PHONE NO.: 231-775-2368 CELL NO.: P.O. NO.:
E-MAIL ADDRESS: lab@cadillac-mi.net QUOTE NO.:

INVOICE TO

CONTACT NAME: [X] SAME
COMPANY:
ADDRESS:
CITY: STATE: ZIP CODE:
PHONE NO.: E-MAIL ADDRESS:

PROJECT NO./NAME: QTRLY BIOSOLIDS
SAMPLER(S) - PLEASE PRINT/SIGN NAME: DAVID SKIFF

TURNAROUND TIME REQUIRED ☐ 1 DAY ☐ 2 DAYS ☐ 3 DAYS [X] STANDARD ☐ OTHER

DELIVERABLES REQUIRED ☐ STD [X] LEVEL II ☐ LEVEL III ☐ LEVEL IV ☐ EDD ☐ OTHER

MATRIX CODE: W=WATER GW=GROUNDWATER WW=WASTEWATER S=SOIL L=LIQUID SD=SOLID SL=SLUDGE DW=DRINKING WATER O=OIL WP=WIPE A=AIR WS=WASTE

ANALYSIS (ATTACH LIST IF MORE SPACE IS REQUIRED)

Certifications: ☐ OHIO VAP ☐ Drinking Water ☐ DoD [X] NPDES
Project Locations: ☐ Detroit ☐ New York ☐ Other

MERIT LAB NO. (FOR LAB USE ONLY)	COLLECTION DATE	COLLECTION TIME	SAMPLE TAG IDENTIFICATION-DESCRIPTION	MATRIX	# OF BOTTLES	NONE	HCl	HNO_3	H_2SO_4	NaOH	MeOH	OTHER	PFAS-MI	Special Instructions
59163.01	2-27-24	1320	EQUIPMENT BLANK	W	3	X							X	
.02	2-27-24	1320	BIOSOLIDS - DIG. 4	SL	4	X							X	ASTM D7968

RELINQUISHED BY: SIGNATURE/ORGANIZATION: C.A. Tomaszewski / City of Cadillac ☐ Sampler	DATE: 2/27	TIME: 1410	RELINQUISHED BY: SIGNATURE/ORGANIZATION: UPS — DATE: 2/28/24 TIME: 1020
RECEIVED BY: SIGNATURE/ORGANIZATION: Ship via UPS Ground	DATE:	TIME:	RECEIVED BY: SIGNATURE/ORGANIZATION: M. Chilcote — DATE: 2/28/24 TIME: 1020
RELINQUISHED BY: SIGNATURE/ORGANIZATION:	DATE:	TIME:	SEAL NO.: SEAL INTACT YES☐ NO☐ INITIALS:
RECEIVED BY: SIGNATURE/ORGANIZATION:	DATE:	TIME:	SEAL NO.: SEAL INTACT YES☐ NO☐ INITIALS:

NOTES: TEMP. ON ARRIVAL 6.0

PLEASE NOTE: SIGNING ACKNOWLEDGES ADHERENCE TO MERIT'S SAMPLE ACCEPTANCE POLICY ON REVERSE SIDE

Rev. 1.26.22